

EXPERIMENT-1 ADDITION

```
Type 'demo()' for some demos, 'help()' for on-line help, or  
'help.start()' for an HTML browser interface to help.  
Type 'q()' to quit R.
```

```
> A=2  
> B=3  
> sum=A+B  
> print(sum)  
[1] 5  
> |
```

EXPERIMENT-2 SUBTRACTION

```
> A=68  
> B=34  
> sub=A-B  
> print(sub)  
[1] 34  
> |
```

EXPERIMENT-3 MULTIPLICATION

```
> a=89
> b=68
> mult=a*b
> print(mult)
[1] 6052
> |
```

EXPERIMENT-4 DIVISION:

```
> a=57
> b=74
> div=a/b
> print(div)
[1] 0.7702703
> |
```

EXPERIMENT-5 ODD OR EVEN

```
Error: unexpected input in "if(a%%2==0){"  
> a <- 45  
>  
> if (a %% 2 == 0) {  
+   print("Even Number")  
+ } else {  
+   print("Odd Number")  
+ }  
[1] "Odd Number"  
> █
```

EXPERIMENT-6 MEAN,MEDIAN,MODE:

```
> # Mean, Median, Mode  
>  
> x <- c(10, 20, 20, 30, 40, 20, 50)  
>  
> # Mean  
> print(mean(x))  
[1] 27.14286  
>  
> # Median  
> print(median(x))  
[1] 20  
>  
> # Mode  
> mode_value <- names(sort(table(x), decreasing = TRUE))[1]  
> print(mode_value)  
[1] "20"  
> █
```

EXPERIMENT-7 SUMMARY:

```
> # Summary of marks
>
> marks <- c(65, 70, 85, 90, 75, 80)
>
> summary(marks)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
65.00  71.25   77.50   77.50  83.75   90.00
> 
```

EXPERIMENT-8 GREATER AMONG THREE NUMBERS:

```
>
>
> # Greatest among three numbers
>
> a <- 20
> b <- 45
> c <- 30
>
> if (a > b && a > c) {
+   print("A is Greatest")
+ } else if (b > a && b > c) {
+   print("B is Greatest")
+ } else {
+   print("C is Greatest")
+ }
[1] "B is Greatest"
> 
```

EXPERIMENT-9 IQR:

```
>
>
> x <- c(10, 20, 30, 40, 50, 60, 70)
>
> Q1 <- quantile(x, 0.25)
> Q3 <- quantile(x, 0.75)
>
> print(Q1)
25%
 25
> print(Q3)
75%
 55
> print(IQR(x))
[1] 30
> █
```

EXPERIMENT-10 QUANTILE:

```
>
>
> # Quantile in R
>
> x <- c(10, 20, 30, 40, 50, 60, 70)
>
> quantile(x)
 0%   25%   50%   75%  100%
 10   25   40   55   70
> █
```

EXPERIMENT-11 MID RANGE:

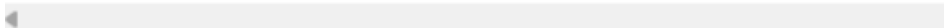
```
>
>
> # Mid-Range in R
>
> x <- c(10, 20, 30, 40, 50)
>
> midrange <- (min(x) + max(x)) / 2
>
> print(midrange)
[1] 30
> █
```

Z-SCOORE NORMALIZATION:

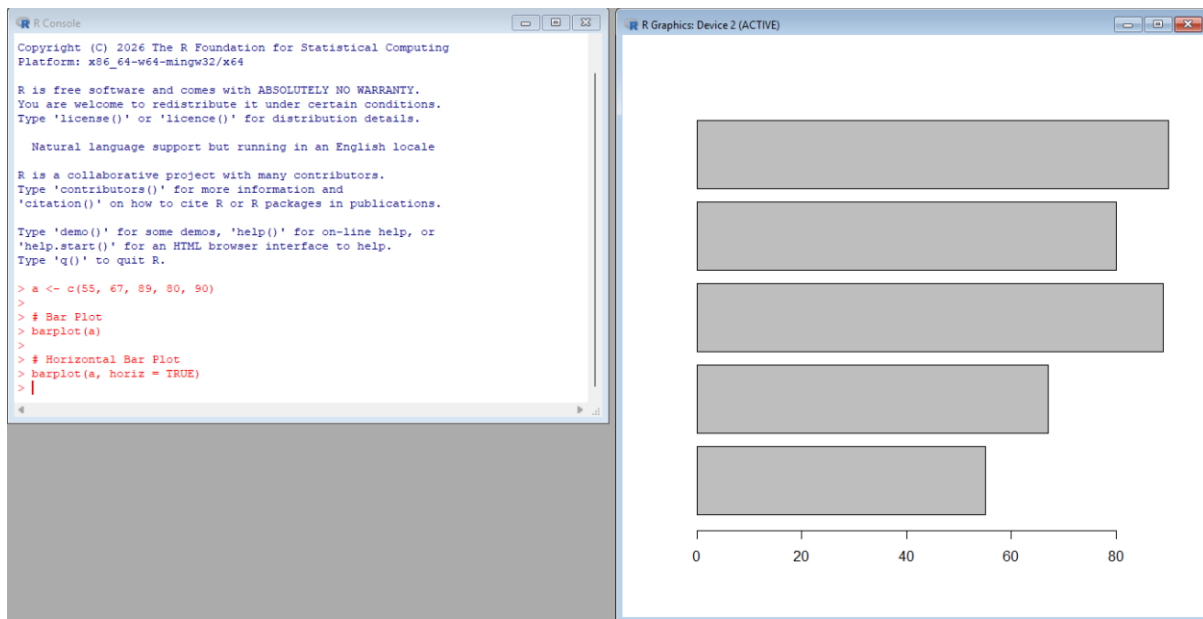
```
> # Input data
> x <- c(10, 20, 30, 40, 50)
>
> # Z-Score Normalization
> z <- (x - mean(x)) / sd(x)
>
> # Display result
> print(z)
[1] -1.2649111 -0.6324555  0.0000000  0.6324555  1.2649111
> █
```

EXPERIMENT-13 MIN,MAX,MEAN,MINMAX:

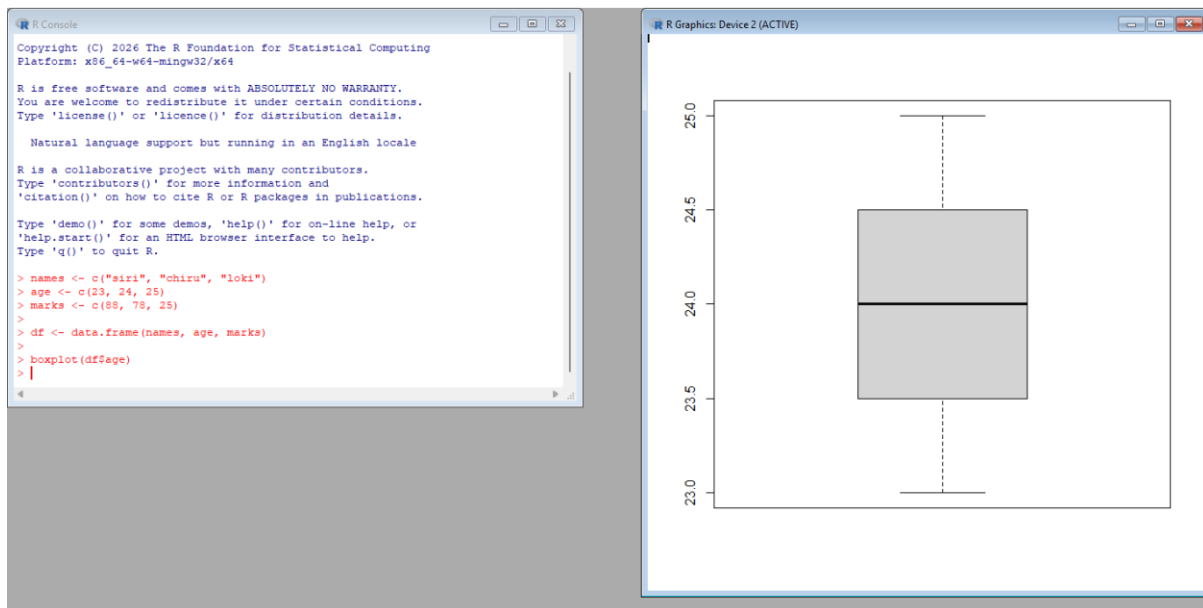
```
>
> # Input data
> x <- c(10, 20, 30, 40, 50)
>
> # Minimum
> print(min(x))
[1] 10
>
> # Maximum
> print(max(x))
[1] 50
>
> # Mean
> print(mean(x))
[1] 30
>
> # Min-Max Normalization
> y <- (x - min(x)) / (max(x) - min(x))
> print(y)
[1] 0.00 0.25 0.50 0.75 1.00
> |
```



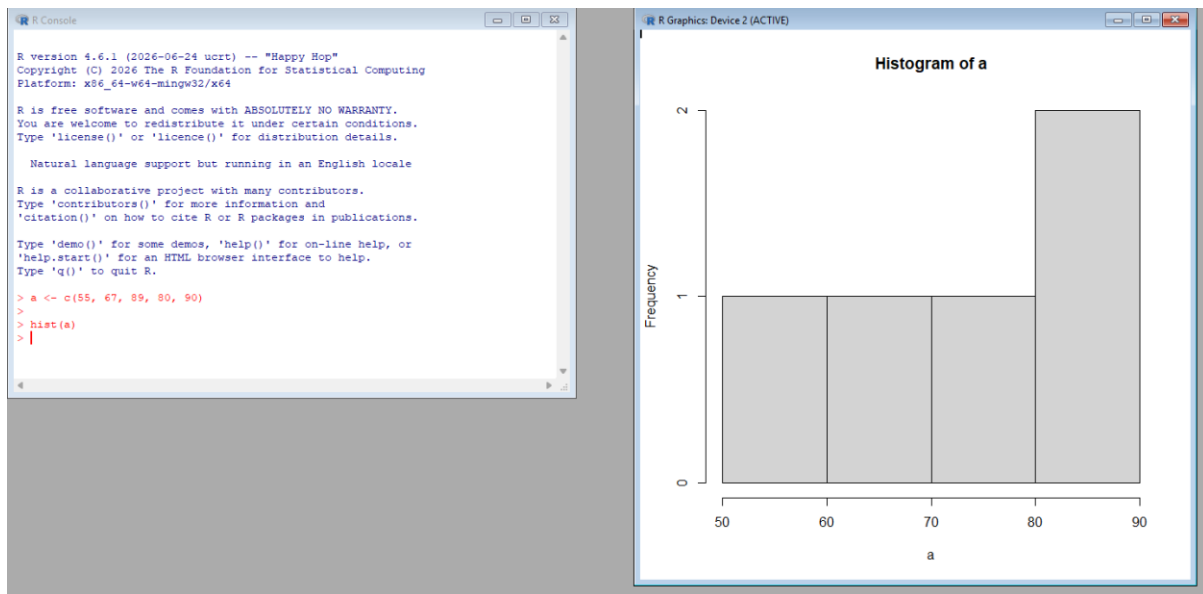
EXPERIMENT 14



EXPERIMENT 15 – BOX PLOT



EXPERIMENT 16 – HISTOGRAM



EXPERIMENT 17 – CORRELATION ANALYSIS

```
R version 4.6.1 (2026-06-24 ucrt) -- "Happy Hop"
Copyright (C) 2026 The R Foundation for Statistical Computing
Platform: x86_64-w64-mingw32/x64
```

```
R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.
```

```
  Natural language support but running in an English locale
```

```
R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.
```

```
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```

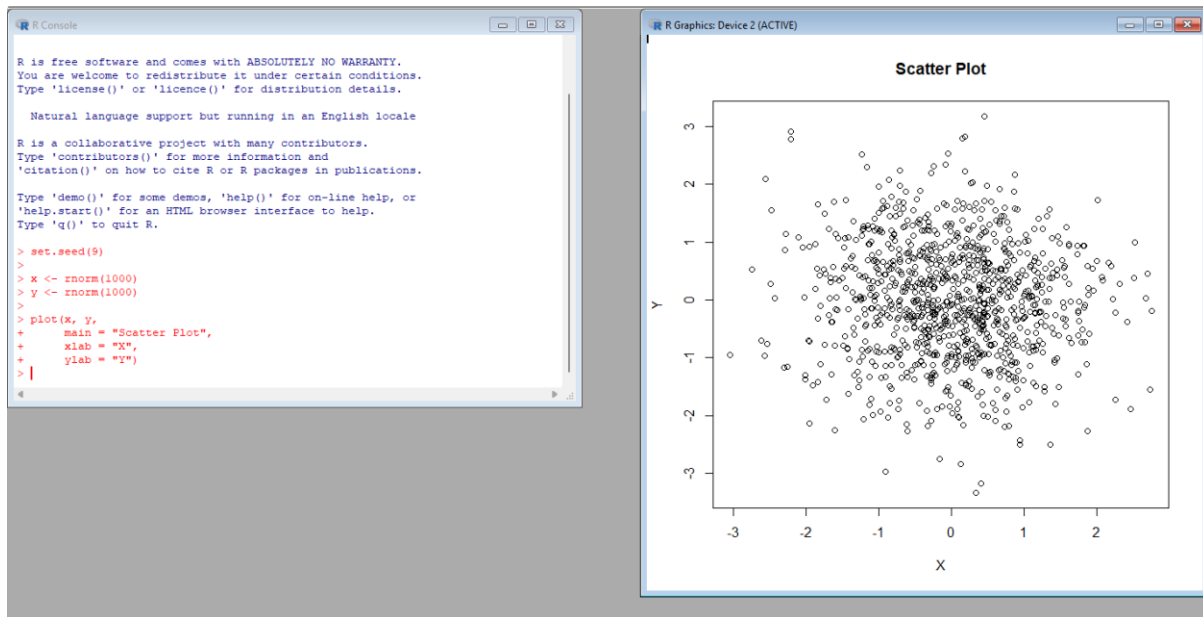
```
> # Sample data
> age <- c(23, 24, 25, 26, 27)
> insulin <- c(80, 90, 85, 100, 95)
>
> # Correlation
> cor(age, insulin)
[1] 0.8
>
> # Correlation test
> cor.test(age, insulin)
```

```
      Pearson's product-moment correlation
```

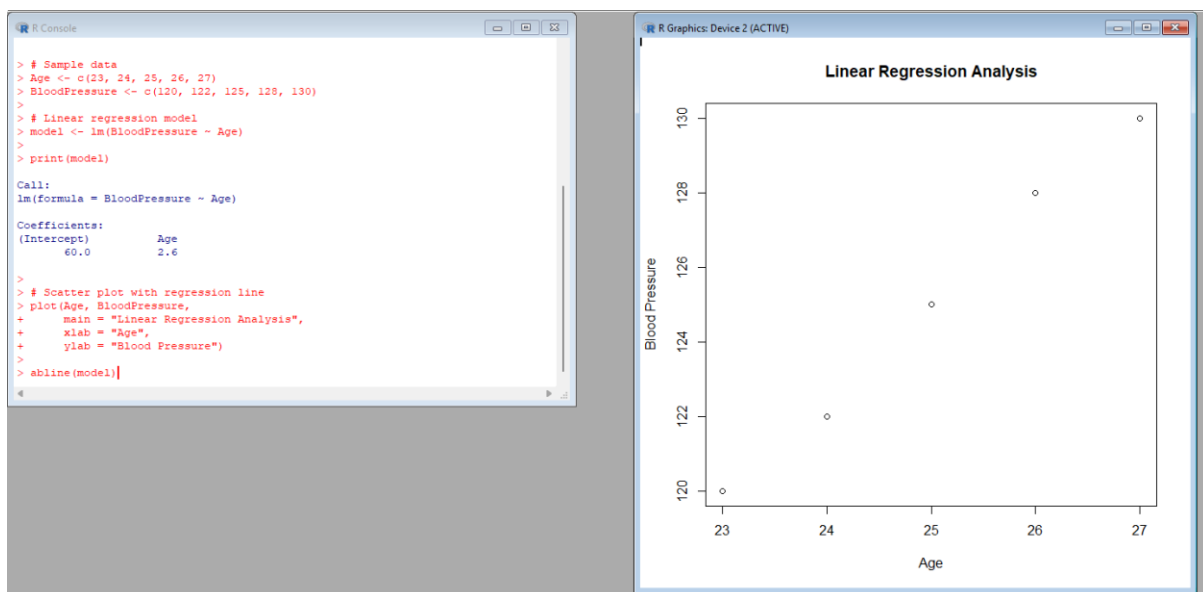
```
data:  age and insulin
t = 2.3094, df = 3, p-value = 0.1041
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.2796400  0.9861962
sample estimates:
cor
0.8
```

```
> |
```

EXPERIMENT 18 – SCATTER PLOT



EXPERIMENT 19 – LINEAR REGRESSION



EXPERIMENT 20 – MULTIPLE REGRESSION

```
> # Sample data
> Age <- c(23, 24, 25, 26, 27)
> BloodPressure <- c(120, 122, 125, 128, 130)
> Glucose <- c(90, 95, 100, 105, 110)
>
> # Create data frame
> input <- data.frame(Age, BloodPressure, Glucose)
>
> # Multiple regression model
> model <- lm(Age ~ BloodPressure + Glucose, data = input)
>
> # Display model
> print(model)

Call:
lm(formula = Age ~ BloodPressure + Glucose, data = input)

Coefficients:
(Intercept)  BloodPressure      Glucose
  5.000e+00    1.831e-15    2.000e-01

>
> # Intercept
> A <- coef(model)[1]
> print(A)
(Intercept)
      5
>
> # Coefficient of BloodPressure
> xBloodPressure <- coef(model)[2]
> print(xBloodPressure)
BloodPressure
1.831027e-15
>
> # Coefficient of Glucose
> yGlucose <- coef(model)[3]
> print(yGlucose)
Glucose
      0.2
>
> # Regression equation
> print(paste("Age =", A, "+", xBloodPressure,
+           " * BloodPressure +", yGlucose, " * Glucose"))
[1] "Age = 4.999999999999988 + 1.8310267194089e-15 * BloodPressure + 0.199999999999999 * Glucose"
>
> |
```